

## **T2' is the absence of a zero-temperature phase transition, and the missing input is closedness of the gapped set**

Define, for the Kogut–Susskind Hamiltonian of the obligations map with gauge group  $G$  and  $N_s$  sites per side,

$$c(g) = \liminf_{N_s \rightarrow \infty} \delta(g; N_s, G), \quad \mathcal{G} = \{g > 0 : c(g) > 0\},$$

where  $\Delta_{a,L} = (\hbar c/a) \delta$ . Then T1 gives  $\delta > 0$  at every finite  $N_s$  and every  $g$ , so a vanishing  $c$  is purely an infinite-volume phenomenon; T2 gives  $[g_0(G), \infty) \subset \mathcal{G}$ ; and T2' is exactly  $\mathcal{G} = (0, \infty)$ . Three statements are proved below. The gap is a continuous function of  $g$  at fixed lattice, by analytic perturbation theory for a simple isolated eigenvalue, so no closure occurs at finite volume. If  $g_* = \inf \mathcal{G} > 0$  then  $c$  vanishes at or just below  $g_*$ , and vanishing of  $c$  admits exactly two mechanisms: infinite-volume vacuum degeneracy, the tunnelling splitting going to zero, or a diverging correlation length. Both are bulk phase transitions at zero temperature, first order and second order respectively, and in the second case the theory acquires a continuum limit at the finite bare coupling  $g_*$ , distinct from the asymptotically free one at  $g = 0$ . Hence

**T2'  $\iff$  4d lattice Yang–Mills has no bulk phase transition at any finite coupling,**

and, on the second-order branch, T2' asserts that the asymptotically free limit is the only continuum limit. For compact  $U(1)$  this fails and the failure is visible in the same terms:  $\mathcal{G}$  is a half-line whose infimum is a second-order point, and the continuum limit there is free Maxwell theory (Guth; Fröhlich–Spencer, abstract level, B78). Since T2 already gives openness of  $\mathcal{G}$  near infinity, the missing property is **closedness**:  $\mathcal{G}$  is open and closed in  $(0, \infty)$  if and only if it is everything. That is the shape of the remaining problem, and the abelian counterexample locates it exactly: the abelian gapped set is open, and closedness is where it fails. Constants explicit; nothing promoted.

### **1. The gap at fixed lattice is continuous in the coupling**

**Proposition 1.** For fixed  $a$ ,  $N_s$  and  $G$ , the map  $g \mapsto \delta(g; N_s, G)$  is continuous on  $(0, \infty)$ , real-analytic off a discrete set, and strictly positive.

*Proof.* Write  $H(g) = \frac{\hbar c}{a} [\frac{g^2}{2} K + \frac{2}{g^2} V]$  with  $K = \sum_{\ell} (-\Delta_{\ell}) \geq 0$  unbounded with compact resolvent and  $0 \leq V \leq 2N|\mathcal{P}|$  bounded, as in the obligations map §1. For every  $g$  in a complex neighbourhood of  $(0, \infty)$  the operators  $H(g)$  share the domain  $D(K)$  and depend on  $g$  holomorphically in the coefficients, so  $\{H(g)\}$  is an analytic family of type (A). By T1 the lowest eigenvalue  $E_0(g)$  is simple and isolated, hence analytic;  $E_1(g)$  is analytic except where it meets  $E_2(g)$ , and in all cases both are continuous by the min-max characterization, since  $\mu_n(g) = \inf_{\dim S=n} \sup_{\psi \in S} \langle \psi, H(g)\psi \rangle / \|\psi\|^2$  is a supremum of functions affine in  $(g^2, g^{-2})$  and therefore continuous. Positivity is T1.  $\square$

So  $\delta(g; N_s) > 0$  for all finite  $N_s$ : the closure of the gap, if it occurs, happens only in the limit  $N_s \rightarrow \infty$ , and  $c$  need not be continuous.

### **2. Two mechanisms for $c(g) = 0$**

Suppose  $c(g) = 0$  for some  $g$ , that is,  $\delta(g; N_s) \rightarrow 0$  along a subsequence of volumes. Let  $\xi(g)$  denote the Euclidean correlation length of the theory at coupling  $g$ , defined as the inverse decay rate of the connected correlator of some local gauge-invariant observable in the infinite-volume limit.

**Proposition 2.** If  $c(g) = 0$  then at least one of the following holds.

1. *Degeneracy.* The infinite-volume theory has more than one ground state: the two lowest finite-volume levels merge with a splitting  $\delta(g; N_s) \rightarrow 0$  while  $\xi(g)$  stays finite.

2. *Diverging length.*  $\xi(g) = \infty$ , that is, some connected correlator decays slower than any exponential.

*Proof.* By reconstruction from the transfer matrix at fixed volume, the connected correlator in the time direction decays with rate  $\Delta_{a,L}/(\hbar c)$  in the vacuum sector, so if the infinite-volume correlation length is finite, say  $\xi < \infty$ , then the infinite-volume theory has a spectral gap  $\hbar c/\xi$  above each of its ground states. Then  $c(g) = 0$  can only come from a splitting between distinct ground states, which is case 1; otherwise  $\xi = \infty$ , which is case 2.  $\square$

Both are bulk phase transitions at zero temperature. Case 1 is first order, with two coexisting vacua and a tunnelling splitting that is exponentially small in the volume; case 2 is second order, with a divergent correlation length. The dichotomy is the standard one, and what matters here is that neither is excluded by anything proved so far.

**Corollary 3.**  $\mathcal{G} = (0, \infty)$  if and only if the theory has no bulk phase transition at any finite coupling. In particular T2' fails if and only if such a transition exists.

### 3. What a second-order point would mean

At a second-order point  $g_*$  the correlation length diverges in lattice units,  $\xi(g)/a \rightarrow \infty$  as  $g \rightarrow g_*$ , so physical masses measured in units of  $1/a$  go to zero and the theory admits a continuum limit taken at  $g \rightarrow g_*$  with  $a \rightarrow 0$  at fixed physical mass. This is a continuum limit taken at the finite bare coupling  $g_*$ , while the asymptotically free limit is taken at  $g \rightarrow 0$  with  $a\Lambda_{\text{lat}}(g) \rightarrow 0$  governed by the two-loop formula of the obligations map §5: two distinct limits of the same lattice theory. Hence, on the second-order branch,

**T2' (second-order branch)**  $\iff$  the asymptotically free limit at  $g = 0$  is the only continuum limit of four-dimensional lattice Yang–Mills.

This is the statement that the lattice theory has a single universality class, believed on the strength of lattice computations and unproven.

### 4. The abelian case seen in the same terms

For compact  $U(1)$  in four dimensions the gapped set is a half-line: a gap at strong coupling (Osterwalder–Seiler, and the  $\nu = 3$  case of the T2 note, which applies verbatim with  $C_2(R_{\min}) = 1$  and  $\dim = 1$ ), and a massless Coulomb phase at weak coupling (Guth, Phys. Rev. D 21 (1980) 2291; Fröhlich–Spencer, Commun. Math. Phys. 83 (1982) 411; abstract level, B78). So  $\mathcal{G}_{U(1)} = (g_c, \infty)$  for some  $g_c > 0$ , which is open and not closed in  $(0, \infty)$ , and the mechanism at  $g_c$  is case 2 of Proposition 2: the photon mass of the confined phase vanishes continuously, and the continuum limit at  $g_c$  is free Maxwell theory.

The abelian theory therefore satisfies every general structural statement above and still fails T2'. Any proof of T2' must use an input that distinguishes the groups. The three places in this programme where that distinction has been made precise are: the commutator potential of G07 §3, which is identically zero for an abelian group; the absence of a gauge-invariant operator linear in the electric field, proved in the upper-bound note Proposition 6, which blocks the abelian gap-closing channel; and the one-loop valley potential of the valley note §3b, which vanishes identically in the abelian theory so that the holonomy is free. All three say the same thing in different variables: the abelian theory has flat directions that nothing lifts.

### 5. Openness, closedness, and where each stands

$\mathcal{G}$  is everything if and only if it is nonempty, open and closed in the connected set  $(0, \infty)$ .

| property                     | status                                      | what it would take                                                                                                       |
|------------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| $\mathcal{G} \neq \emptyset$ | proved: $[g_0, \infty) \subset \mathcal{G}$ | the T2 note                                                                                                              |
| $\mathcal{G}$ open           | open in general; true near $\infty$         | a volume-uniform stability theorem: a gap at $g$ survives a small change of $g$ with constants independent of the volume |
| $\mathcal{G}$ closed         | <b>the missing input</b> ; false for $U(1)$ | exclusion of both mechanisms of Proposition 2 at every finite coupling                                                   |

Openness is the kind of statement that stability theory supplies for gapped local Hamiltonians when a Lieb–Robinson bound is available; for the Kogut–Susskind Hamiltonian the electric term is unbounded, so the standard bounds require an extension, which makes openness a bounded technical question. Closedness is the conjecture itself. Recording the split is useful because it separates a technical obstacle from the mathematical content, and because it shows that a proof cannot proceed by continuity alone: the abelian theory has an open gapped set too.

## 6. Consequence for STATE

The lower side now has a precise formulation: T2' is closedness of  $\mathcal{G}$ , equivalently the absence of a zero-temperature bulk phase transition, equivalently, on its second-order branch, the uniqueness of the continuum limit. The two sub-targets this names are a volume-uniform stability theorem, which needs a Lieb–Robinson bound for a Hamiltonian with unbounded electric terms, and the exclusion of vacuum degeneracy and of a diverging correlation length. The first is technical and bounded; the second is the conjecture. Between them, the stability question is the only one of the two that present methods can be expected to settle, and it is the natural next step on this side.